Terrain-Conditioned Diffusion for Generating Street-Level Expansion Layouts of Small Towns

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Code: https://github.com/eshankiyer/terrain-urban-diffusion

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Abstract

Generative models for urban layouts have concentrated on large, grid-dominated cities, typically in the United States, and almost universally ignore the terrain the city sits on. Yet for the small towns that make up most settlements worldwide, and especially for towns in hilly regions of Europe and Asia, topography is a first-order constraint on where streets and buildings can go. We present a conditional denoising diffusion model that generates street-level expansion layouts (road rasters and continuous built-up density) for small towns, conditioned jointly on terrain (elevation and slope) and the observed settlement state. The model is trained on windows around approximately sixty towns in hilly terrain across Europe and Asia, assembled automatically from OpenStreetMap vectors, open elevation tiles, and the multi-epoch GHS-BUILT-S built-up surface, whose 1980–2020 epochs provide real temporal supervision: each sample conditions on the town as observed at one epoch and targets the next. A concentric-erosion proxy that manufactures pseudo-temporal pairs from a single snapshot, used in an earlier version of this work, is retained as a baseline. An eleven-metric raster scorecard (15-minute amenity access, infill, land efficiency, hazard avoidance, green preservation, access equity, and related measures) turns the sampler into an optimizer by best-of-N selection, and post-hoc classifiers annotate selected plans with bike infrastructure and typed zones. Training converges from 0.744 to approximately 0.04 noise-prediction loss over 120 epochs; we show top-ranked generated futures for a held-out town with per-pixel walk-time maps, specify the held-out evaluation protocol, and release code, the town list, and the full pipeline. A single trained model supports proposal generation for real towns, iterative sandbox growth on arbitrary heightmaps, and continuation of user-drawn plans.

This version (v4) extends the pipeline in four directions and reports what actually happened when we retrained on a mixed dataset. First, we lower the growth-pair threshold and add peri-urban fringe windows, which more than doubles the training pairs; the density and amenity channels come out spatially coherent and conditioning-faithful, while the road channel, still supervised at 1-pixel width, remains unlearnable at this data scale, a negative result reported as such. Second, we add a fifth conditioning channel for water and zero growth targets on it, because the deployed 4-channel model was building on lakes; a RePaint-style hard lock on water and slopes above 25° patches this at inference time without retraining. Third, we add a two-expert mixture (hilly towns vs. flat metro fringes, including new US windows) with a hard, auditable router on conditioning statistics rather than a learned gate, and report it as architecture merged with training still pending. Fourth, we add a planner-facing tool layer (region-locked inpainting, multi-sample consensus maps, GeoJSON vector export, and a natural-language operations layer that maps planner intent to a validated, geometry-free operation schema executed against those tools), together with an updated zoning stage and a browser deployment that runs the diffusion model and the zone classifier client-side.

The present version (v5) extends the pipeline in two further directions. First, the two-expert mixture is widened to five experts covering the principal contemporary growth regimes: village, hilly small town, planned flat fringe, informal peri-urban frontier, and megacity edge. Dispatch remains a hard, auditable rule on conditioning statistics, extended with a roads-per-built-area ratio that separates unplanned from planned growth, and 84 additional training and evaluation windows across Latin America, Africa, South and Southeast Asia, China, Australasia, and Canada accompany the new experts, with a Delhi-region window held out for evaluation. Second, we diagnose and correct a failure of the typed-zone classifier observed on a car-dependent United States suburb, where a majority of candidate cells were labelled institutional: the cause is per-window maximum normalisation of the amenity-density feature, which we replace with an absolute calibration, complemented by per-class abstention margins. All five experts trained to convergence, and the reproduction protocol yielded a negative result of independent interest: with the expanded training distribution in place, the original failure does not reproduce under either normalisation, indicating that the training data, rather than the feature scale, was the operative correction. The per-class margin behaves monotonically and is retained as the control for institutional assignments.

1 Introduction

Most generative work on urban form asks the question: what does a city look like? This paper asks a narrower and, we argue, more practically useful question: given a particular small town, on particular terrain, what would its next increment of growth plausibly look like?

The distinction matters for three reasons. First, most of the world's settlements are not large cities. Small towns dominate by count, and their growth decisions, such as where a new residential street goes or which hillside is terraced, are made with far less planning capacity than a metropolitan government has. A tool that proposes terrain-respecting expansion options is more useful at this scale, not less. Second, terrain is the binding constraint precisely where planning capacity is lowest: the hill towns of Nepal, the Italian Apennines, or the Western Ghats cannot borrow street patterns from Chicago. Third, existing generative models have a documented geographic bias: the most capable urban layout generators are trained on large, mostly American cities [2, 3, 4], whose gridded morphology transfers poorly to organically grown towns; where non-US cities have been included, the models have measurably struggled with non-grid street orientation [5].

Terrain-aware settlement growth modelling is not itself new. The SLEUTH family of cellular automata has used slope as a growth driver since the 1990s [1], and modern deep-learning land-use-change models routinely include elevation among their covariates [6]. But these models predict growth *probability* at coarse resolution: their output is a per-pixel likelihood of urbanisation, not a design. They cannot say where the street goes. Conversely, recent diffusion-based layout generators produce street-level designs but condition on land use, text, or nothing at all, but not on topography [2, 7, 8]. Diffusion models that do engage with terrain generate the terrain itself [9, 10], not settlements upon it. The specific combination this paper addresses, namely street-level generative layout design conditioned on terrain for small towns and trained on European and Asian data, is to our knowledge unoccupied.

Our contributions are:

1. A fully automatic data pipeline that assembles terrain-conditioned growth training pairs for arbitrary towns from OpenStreetMap, open elevation tiles, and the multi-epoch GHS-BUILT-S built-up surface [21], requiring no manual annotation and no API keys, together with a curated list of ∼60 training and 8 held-out small towns in hilly terrain across Europe and Asia. Supervision is real temporal change between observed epochs (1980–2020); the concentric-erosion proxy of the first version of this work, which recovers pseudo-temporal

pairs from a single snapshot, is kept as a baseline and ablation, and we state the temporal proxy that remains (roads) explicitly.

2. A compact (∼13M parameter) conditional diffusion model that generates joint road/built-density growth rasters, where density is continuous so that infill densification and greenfield sprawl are both supervised and both generable.

3. An eleven-metric sustainability scorecard, computed directly on rasters and the extracted road graph against real OSM amenities, greenspace, and terrain hazard proxies, that turns the sampler into an optimizer via best-of-N selection without retraining.

4. Post-hoc annotation stages, a slope-aware bike-infrastructure edge classifier and a typed-zone classifier, that add planning detail the diffusion targets cannot carry, each with a leave-one-town-out transfer protocol.

5. A demonstration that one trained model supports proposal generation for real towns, iterative game-like growth of synthetic settlements on arbitrary heightmaps, and (in the v3 extension) continuation of user-drawn plans, which we discuss as the decision-making layer of a prospective automated city-building agent.

2 Related Work

2.1 Generative urban layout models

Procedural methods based on shape grammars and production rules have long been used to synthesise street networks [11, 12], and remain the standard in game tooling; their weakness is the manual rule-crafting required per style. Learning-based approaches replaced rules with data: Neural Turtle Graphics models road layouts autoregressively [13], GlobalMapper generates building layouts for arbitrary blocks with graph attention [4], and CityGen [3] and CityDreamer [14] produce unbounded city layouts with diffusion. Closest to our setting, He et al. propose a stepwise multimodal-diffusion urban design framework trained on Chicago and New York [2], and explicitly note the restriction to US-grid morphology as a limitation. GAN-based urban form models trained on multiple cities report degraded fidelity on non-grid cities [5]. None of these condition on terrain.

2.2 Urban growth and land-use-change models

SLEUTH [1] takes slope, land use, exclusions, urban extent, transport and hillshade as inputs to a cellular automaton and remains widely applied [15]; extensions couple CA with support vector machines [16] or deep networks [6]. Deep sequence models (ConvLSTM) forecast urban growth from satellite time series [17]. These models incorporate terrain but output coarse urbanisation probabilities rather than street-level layouts; they simulate *whether* a cell urbanises, not *what is built*. Our temporal supervision draws on the same class of multi-epoch built-up products these models consume [21], but uses them as diffusion targets rather than CA calibration data.

2.3 Terrain and geospatial generation

Diffusion models generate realistic terrain from text or sketches [9, 10, 24], and map-or-primitive-conditioned diffusion synthesises satellite imagery [7, 22]. GPS coordinates have been used as conditioning for image generation [23]. This line of work treats terrain as the *output* or the imagery as the target; we invert the relationship, treating terrain as a constraint under which settlement structure must be generated.

2.4 Generative models as game agents

Automating construction in city-building games has mostly meant procedural map generation. Playing such a game requires perception, decision-making, and actuation layers. We position our model as the decision-making layer (current city state and heightmap in, next construction increment out) and discuss integration via game modding APIs in Section 10.

3 Method

3.1 Problem formulation

Let $E \in \mathbb{R}^{H\times W}$ be an elevation raster over a window of $\sim$1.9 km (H=W=128 at 15 m/px), S its slope, $D_t \in [0,1]^{H\times W}$ the observed built-up density at epoch t (the built fraction of each cell), and R_t a binary raster of the road network inside the epoch-t settlement footprint. The model learns the conditional distribution $p(R_{\text{new}}, D_{t+1} \mid E, S, D_t, R_t)$ over roads outside the epoch-t footprint and the next-epoch density field. Conditioning channels are concatenated: elevation is z-scored per window and clipped to $\pm 3\sigma$; slope is normalised by 30°; density is used directly. Targets are affinely mapped to $[-1,1]$ (binary road targets to $\{-1,1\}$). Because density is continuous rather than a binary footprint, the target expresses both infill densification (cells that were already built getting denser) and greenfield sprawl (new cells), and both are supervised.

3.2 Temporal supervision from multi-epoch built-up data

The primary training regime replaces manufactured growth pairs with observed ones. GHS-BUILT-S R2023A [21] provides a global built-up surface at 3 arc-seconds on the WGS84 tile grid for the epochs 1980, 1990, 2000, 2010, and 2020; our pipeline downloads and caches the relevant 10° tiles once and performs windowed reads per town. For each town and each consecutive epoch pair $(t, t{+}1)$ with sufficient observed growth we emit one sample: the model conditions on elevation, slope, the observed density D_t, and roads clipped to the epoch-t footprint, and targets D_{t+1} together with roads outside that footprint. Windows whose epoch pair shows too little change are dropped automatically.

One proxy remains. OpenStreetMap carries no road history for these towns, so "roads at epoch t" are necessarily present-day OSM roads restricted to where density was already non-zero at epoch t. Roads built later inside an old footprint therefore leak into early-epoch conditioning; we flag the consequences in Section 11. The built-up channels, which carry the growth signal itself, are genuinely temporal.

3.3 The concentric-erosion proxy (v1 baseline)

The first version of this work constructed supervision from present-day snapshots alone: the observed built-up mask B was eroded with a disc of radius $\rho = 10$ px ($\approx$150 m) to produce a pseudo-historical core B_c, and the ring $B_r = B \setminus B_c$ became the generation target, with ring roads defined analogously. The proxy assumes predominantly outward, concentric growth. That assumption is reasonable for compact hill towns whose steep surroundings suppress leapfrog development, wrong for linear valley towns and for growth dominated by infill, and structurally incapable of expressing densification, since its targets are binary. We retain it as a baseline and ablation: the channel counts match the temporal regime, so the same architecture trains on either supervision unchanged, which is precisely what makes the comparison clean.

3.4 Architecture and training

A U-Net with three resolution levels (64/128/256 channels), residual blocks, sinusoidal timestep embeddings, and self-attention at the two coarsest resolutions predicts the added noise ϵ under a cosine noise schedule with $T{=}1000$ steps [18, 19]. The 4 conditioning channels are concatenated to the noisy target channels at the input ($\sim$13M parameters). We train with AdamW, cosine learning-rate decay, gradient clipping at 1.0, and an EMA of the weights (decay 0.999); sampling uses 50 DDIM steps [20]. Each town window is augmented by the dihedral group (8 orientations) and small centre jitter.

3.5 Inference modes

Real mode. For a real town, the pipeline fetches current elevation, OSM rasters, and the latest GHSL epoch, and samples a next-epoch density and road increment conditioned on the true terrain and observed state. **Sandbox mode.** For an arbitrary heightmap (fractal noise here; a game-exported heightmap in principle), a seed settlement of a few pixels is placed on low-slope ground and the model is applied iteratively: the union of the previous state and the generated increment becomes the next conditioning state. Each round is one "turn" of automated play. The plan-import mode of Section 6 generalises this by letting a human drawing, rather than a seed, initialise the state.

4 Sustainability evaluation and best-of-N selection

A generative model of plausible growth is not yet a planning tool: a planner needs to know which of many plausible futures is *better*. We therefore attach an eleven-metric raster scorecard that grades any (roads, density) plan against its terrain and environmental context, and use it for best-of-N selection: sample N futures from the diffusion model, score each, keep the top-ranked ones. This turns the sampler into an optimizer without retraining, and the same scorecard grades real draft plans for critique.

The context layers are fetched once per town: real OSM amenities in seven categories (food, education, health, civic, recreation, transit, social), greenspace and water masks, and two terrain hazard proxies derived from the DEM. The metrics, each normalised to $[0, 1]$, are:

- **15-minute amenity coverage** (weight 0.18): per-pixel network walk times are computed by Dijkstra over road pixels at 80 m/min with a 15-minute (1200 m) budget, with real OSM amenities as fixed destinations; coverage is the built-weighted served fraction over the amenity categories *present in the window*, so small towns are compared on what exists. A category that exists but is unreachable over the network contributes zero while remaining in the denominator.
- **Infill share** (0.10): fraction of new growth inside the existing footprint rather than greenfield.
- **Land efficiency** (0.08): added density per newly consumed pixel.
- **Circuity** (0.06): network over straight-line distance between sampled road-pixel pairs.

- **Earthwork** (0.06): mean terrain slope under *new* roads, a grading-cost and erosion-risk proxy.
- **Green preservation** (0.10): survival of *baseline* functional green patches ($\geq$0.45 ha); new green elsewhere cannot buy back cleared green, and new roads clear green just as buildings do.
- **Green access** (0.08): share of built pixels within a 300 m network walk of a remaining functional green patch.
- **Flood avoidance** (0.10): new development on flood-prone cells, flagged by a HAND proxy: height above the nearest mapped water under 5 m within $\sim$105 m of water.
- **Landslide avoidance** (0.08): new development on slopes above 25°, with slopes above 35° counted double.
- **Congestion proxy** (0.08): one minus the Gini coefficient of edge betweenness on the road graph extracted from the raster, so plans that funnel all traffic through one link score low.
- **Access equity** (0.08): population-weighted dispersion of per-cell amenity coverage, penalising plans that serve the centre and strand the periphery.

Weights sum to one. Hazard and earthwork terms are computed over new development only, so a town's pre-existing switchbacks are not held against a plan. Ranking applies a growth-delivery factor, so a do-nothing plan cannot win best-of-N simply by avoiding every hazard: the scorecard rewards delivering growth well, not refusing to grow.

Two boundaries are deliberate. The metrics are spatial-form proxies computed from rasters; HAND and slope thresholds are not hydraulic or geotechnical models, and no score certifies environmental performance. Socioeconomic factors (income, tenure, affordability, displacement risk) are named as required human inputs and are deliberately not estimated from rasters. As a methods-transparency note: the metric design came from a multi-perspective specification (ecologist, transport engineer, geotechnical engineer, equity planner) and the implementation survived a two-round adversarial verification loop that caught one critical defect, pre-existing roads scored as new development, before acceptance.

5 Post-hoc annotation stages

Two planning layers are kept out of the diffusion targets because their OSM labels are too sparse in the training towns to supervise generation; both are instead trained where labels are dense and applied to generated rasters, with leave-one-town-out protocols so the transfer claim is testable rather than asserted.

5.1 Bike-infrastructure edge classification

Cycleway tagging is dense in the flat cycling countries and nearly absent in mountain towns, so bike lanes are a per-edge classifier, not a generated channel. We train a logistic classifier on twelve well-tagged flat European towns (Dutch, Danish, and north-German), using physical features that transfer across morphology: mean absolute grade along the edge computed from the DEM, road class, edge length, and the edge's position relative to the town centre. Evaluation is leave-one-town-out. At inference, a skeleton-to-graph extractor converts any generated road raster into junction-to-junction polylines, and every edge receives a bike-infrastructure probability, so a plan selected in Section 4 arrives annotated with the edges where protected cycling infrastructure is physically plausible.

5.2 Typed zones

The diffusion model generates one undifferentiated density field; a planner needs to know where residential, commercial, industrial, and institutional uses go. OSM landuse polygons provide labels, but they are patchy outside Europe and heavily imbalanced toward residential, so zoning is likewise a post-hoc per-pixel classifier trained only where labels exist, on features the generator already produces (distance to centre, slope, road proximity, local density, amenity density, local road density), with per-class caps against residential dominance. Evaluation is leave-one-town-out macro-F1; at inference the classifier paints use classes onto new development.

6 Extensions in progress (v3)

Three extensions are implemented in the released pipeline; at the time of writing they were training and not yet reported. Section 8 gives the retraining result for the amenity channel and the mixed dataset it trained on. Plan import and plan-style rendering remain method-only in this version too, so we still describe all three below as method first.

Generated amenity channel. The v2 scorecard measures access against *existing* amenities only, which correctly penalises remote growth but cannot express "grow here and put a school there." v3 adds a third target channel of log-scaled amenity density rasterised from OSM points, so the model learns where local centres sit relative to fabric and, at sampling time, *proposes* amenity locations for new growth. The scorecard reports access both with and without the proposed amenities, so the assumption remains visible in the reported scores.

Plan import. A user-drawn plan image following a simple convention (black lines for roads, warm colours for built area, green kept free) is parsed into conditioning rasters and continued by the model, over real terrain, a supplied heightmap, or procedural relief. This generalises sandbox mode: the initial state comes from a human drawing instead of a seed pixel, and the conditioning layout is unchanged, so any v2/v3 checkpoint works.

Plan-style rendering. Raw model rasters are hard for humans to read, so a rendering stage composes hillshaded terrain, roads, zone colours, green and water, and proposed local centres into a planning-map figure with a legend and scale bar.

7 Experiments

7.1 Data

$\sim$60 training towns and 8 held-out towns (population roughly 5k–60k) in hilly regions of Europe (Alps, Apennines, Iberia, Balkans) and Asia (Japan, the Indian Himalaya and Western Ghats, Nepal, Vietnam, Indonesia, Turkey, the Caucasus), listed in the repository. Each town contributes up to four consecutive GHSL epoch pairs (1980$\rightarrow$1990 through 2010$\rightarrow$2020); pairs with insufficient observed growth are dropped automatically, and each surviving window is expanded eightfold by dihedral augmentation. The exact per-run sample manifest is produced by the released pipeline.

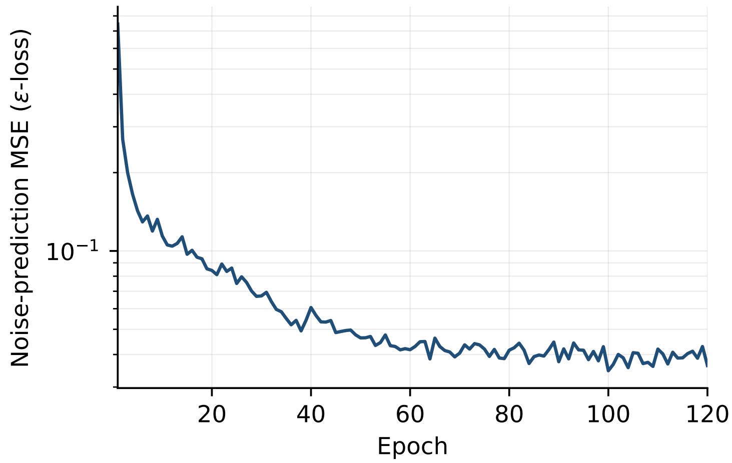

Figure 1: Training loss of the v2 model (noise-prediction MSE, log scale) over 120 epochs on the GHSL temporal pairs: 0.744 at epoch 1 to ∼0.040 at epoch 120.

7.2 Evaluation protocol

On each held-out town the protocol compares the generated increment to the real next-epoch change: **Slope KS**: Kolmogorov–Smirnov distance between the slope distributions under generated vs. real new built-up pixels (lower is better; measures whether the model builds on the kinds of slopes the town actually built on); **Road connectivity**: fraction of generated road pixels 8-connected to the existing network; **Built contiguity**: fraction of new built pixels adjacent to at least two settlement pixels; **Volume ratio**: generated/real new built area; plus ordinal density-class agreement derived by binning the continuous density channel. Two comparisons are planned: a slope-blind baseline that scatters the same built mass uniformly at random over the window (a deliberately weak control that isolates whether the model uses terrain at all), and the erosion-trained v1 model as an ablation of the temporal supervision. The protocol is implemented in the released pipeline; we do not report held-out numbers in this version because that evaluation run has not completed. For the post-hoc stages, the corresponding protocols are leave-one-town-out AUC (bike edges) and leave-one-town-out macro-F1 (zones), likewise shipped with the code.

7.3 Results

Training. The v2 temporal regime trains stably: the noise-prediction MSE falls from 0.744 at the first epoch to ∼0.040 by epoch 120 (final-epoch 0.036, mean of the last ten epochs 0.039), with no divergence and the familiar fast-then-slow diffusion profile (Figure 1). We read this only as evidence that the continuous-density temporal targets are learnable by the unchanged v1 architecture; training loss is not a fidelity claim.

Qualitative selection results. Figure 2 shows the pipeline end to end on a held-out town: many futures are sampled, the scorecard of Section 4 ranks them, and the top three are shown with their per-pixel walk-time maps and bike-lane annotations. The three selected futures score within one point of each other (51, 50, 50 of 100) while proposing visibly different spatial arrangements, and their 15-minute amenity coverage of built pixels ranges from 82% to 87%. This is the intended behaviour of best-of-N selection: several near-equivalent options for a human to choose among rather than a single answer. The density fields are speckled at the pixel level, as raw diffusion rasters are; the rendering stage of Section 6 exists precisely because these rasters are for scoring, not for reading.

8 v4: mixed-dataset retraining, water conditioning, mixture of experts, and planner tooling

Section 6 described three v3 extensions as method, with results deferred. This section reports what happened when we retrained on the extended data, adds a water-conditioning channel the v3 model lacked, introduces a two-expert mixture, and adds a set of planner-facing tools on top of the trained model. Throughout this section, results are reported as observed, including negative findings.

8.1 Mixed-dataset retraining

The v2/v3 growth-pair filter of Section 3.2 required at least 60 px of observed change between consecutive GHSL epochs. In practice this discarded most slow-growing towns, leaving too few pairs to train on. We lower the threshold to 30 px: a slow-growing town still provides supervision for infill behaviour even where it provides little else. We also add 31 peri-urban fringe windows on the growth edges of mid-size cities (Section 8.3 gives the list), since most contemporary settlement growth is metro-fringe absorption, not village expansion, and the original town list had none of it. Together these changes take the training set from 59 real epoch-pair windows and 36 usable growth pairs to 472 augmented samples after dihedral augmentation, up from 288.

Not all output channels learned equally. The density channel and the amenity channel both come out spatially coherent: a pixel's value predicts its neighbours' values, with neighbour correlation around 0.98, and the generated patterns track the conditioning (terrain, existing density, existing amenities) closely enough to read as a plan rather than noise. The road channel does not reach the same place. Roads are supervised as a 1-pixel-wide line on a 128x128 grid, and at this data scale the model has not learned a road field with the same coherence: measured the same way, neighbour correlation on the road channel sits around 0.26, close to what an unstructured field would give. The deployed site (Section 8.7) hides the generated road channel and falls back to the existing OSM network for anything that needs roads, which is a real limitation on what the model can currently propose, not a cosmetic choice. The fix we have identified but not yet run is a retrain with the road target dilated to several pixels of width before supervision, which should give the loss a wider basin to learn a coherent line rather than a sparse near-random one; we state this as a planned fix, not a result.

A further correction concerned the display calibration rather than the model. Our first rendering pass classified a pixel as "new growth" once its decoded density exceeded 0.12, and this consistently overstated growth extent against the training distribution; recalibrating the growth threshold to 0.25 (used throughout the render and consensus tooling described below) brings displayed growth back in line with what the density channel actually predicts. This is a display-calibration fix, not a retraining fix: the underlying density channel is unchanged, only the threshold used to decide which pixels count as "built" for rendering and for the consensus and zoning stages that read the density field.

8.2 Water conditioning

The v2/v3 model conditions on 4 channels (elevation, slope, density, roads) and had no representation of water at all. On flat ground, a lake and a buildable valley floor look the same to those 4 channels, and the deployed model would occasionally generate growth directly onto mapped lakes.

v4 adds a fifth conditioning channel: a filled water mask rasterised from OSM water and wetland polygons (lakes, wetland, riverbanks, docks). The three target channels are unchanged in count, but growth is forced to zero on water pixels during training, so the model is taught that water pixels do not urbanise rather than left to infer it from a channel that cannot express it. This changes the checkpoint format: a 5-channel v4 dataset is not compatible with a 4-channel v3 checkpoint, and we keep the two cache directories separate so a stale 4-channel cache cannot be silently reused.

Retraining the water-conditioned model is not yet complete, so in the interim we ship an inference-time fix that requires no retraining at all: a RePaint-style hard lock (Section 8.4) applied to water pixels and to slopes above 25°, the same landslide-prone threshold used by the sustainability scorecard (Section 4). Locked pixels are held at their known (non-buildable) state through every reverse diffusion step and composited back exactly at the end, so the model is structurally prevented from building on water or on steep ground regardless of what the 4-channel conditioning tempts it to do. This is a deployment patch, not evidence that a future water-conditioned retrain is unnecessary; the two are complementary; the lock is retrain-free and ships now, the conditioning channel is the fix that lets the model learn the constraint rather than have it imposed from outside.

8.3 Mixture of experts

Training one model on hilly small towns and flat metro fringes together asks it to represent two different growth regimes: hill towns with sparse, curved, slope-constrained street networks, and metro fringes with dense gridded or subdivision fabric on nearly flat ground. v4 splits this into two experts. The town expert keeps the original hilly small-town data. The urban expert trains on flat metro-fringe windows, combining 31 peri-urban fringe windows across Europe, Asia, Africa, and the Americas with 20 new US metro-fringe windows (Midwest, South, and West growth frontiers, chosen for gridiron and subdivision morphology on flat or gently rolling terrain), giving the urban expert its first American training data.

Routing between the two experts is a hard rule on conditioning statistics, not a learned gate: the mean of the slope channel, the built fraction, and the road-pixel fraction of a window each cast one vote for "urban," and a window routes to the urban expert on a 2-of-3 majority, otherwise to the town expert. With only two experts, we chose interpretability over a learned gate deliberately: a hard vote on named statistics can be audited and logged (which threshold fired, by how much), and it cannot silently misroute the way a learned gate can fail open. Batched dispatch preserves input order, so a mixed batch is split by vote, sampled per expert, and reassembled without changing which output corresponds to which input window. The cost is that a window near a threshold boundary is assigned entirely to one expert with no blending.

As of this writing the mixture-of-experts architecture and both training datasets are merged into the repository; training the urban expert to convergence has not run. We report the design and the data, not a result, and we do not claim the two-expert model outperforms the single mixed model of Section 8.1 until that training run completes. Section 9.2 supersedes this two-expert design with a five-expert mixture; the routing philosophy is unchanged.

8.4 Planner-facing tooling

The trained diffusion model produces one raster future at a time. This subsection adds four tools that turn that model into something closer to a planning instrument, all of them training-free additions on top of the existing checkpoint.

Region-locked inpainting. A RePaint-style sampler lets a planner mark part of the window as locked (existing fabric, floodplain, protected land) and regenerates only the free region. At each reverse diffusion step the locked region is replaced by the known content forward-diffused to the current noise level, so the model always denoises against consistent locked context; periodic resampling jumps (rewinding several steps and redoing them) let the free region re-harmonise with the locked boundary instead of leaving a visible seam. Locked pixels are hard-composited at the end and are therefore exact by construction; this is what makes the water and steep-slope lock of Section 8.2 possible without retraining. A caveat carries over from that section: the model was never trained with masked context, so a hard lock is a distribution shift, and a narrow free region squeezed between locked fabric can come out blander than an unconstrained sample.

Multi-sample consensus maps. Sampling many futures under identical conditioning and aggregating them per pixel turns one arbitrary draw into a share-of-futures-built map: a pixel that is built in 87% of sampled futures is a stronger proposal than one built in 30%. Pixels in the 30–70% agreement band are reported separately as contested frontiers, the places the model itself cannot decide on. This is the cheapest way to turn an image generator into a planning tool, the resulting agreement rate is a statistic of the model's learned distribution, not a calibrated probability of future development; the road channel is excluded from consensus statistics for the reason given in Section 8.1.

GeoJSON vector export. Generated road, zone, and density rasters are only useful to us; a working planner needs vectors in a GIS tool. Roads are skeletonised and converted to junction-to-junction polylines (reusing the bike-lane stage's skeleton-to-graph extractor), with a second pass that recovers isolated loops the first pass misses, and simplified with Douglas-Peucker in pixel units. Zone and growth rasters are converted to polygons by connected-component labelling and polygon union rather than contour tracing, which costs speed but handles holes correctly. Everything is written as one GeoJSON FeatureCollection in WGS84 using the same window georeferencing as the data pipeline, openable directly in QGIS or ArcGIS. Positional accuracy is bounded by the 15 m raster grid; these are sketches for planning discussion rather than survey-grade vectors, and the export metadata states this.

Natural-language operations layer. A planner can describe intent in a sentence rather than drawing a mask or tuning scorecard weights. A parser (a keyword-matching rule parser, or optionally a small local LLM served through Ollama, with the rule parser as an automatic fallback on any failure) maps a request onto a short list of verified operations: Regenerate a region, SetZoneBias for a zone class in a region, AddAmenityTarget for a category in a region, Protect a region, or Rerank the scorecard weights. Regions are limited to a fixed vocabulary (north, south, east, west, centre, edge, floodplain, steep, riverside) resolved to a mask by simple, auditable rules, never by anything the parser invents; zone classes and scorecard metrics are checked against closed vocabularies before anything executes, and a clause the parser cannot map becomes a warning, never a guessed operation. Every operation routes onto tools that already exist: Regenerate and Protect become the inpainting lock above, AddAmenityTarget biases the amenity channel toward a region, SetZoneBias reweights the zone classifier's per-class probabilities inside a region, and Rerank recomputes best-of-N ordering under new scorecard weights. The result of every one of these operations still passes through the same eleven-metric scorecard of Section 4 before it is shown, so a request that would pave the floodplain gets a visibly low score rather than being executed and left uncritiqued. The language layer routes and validates; it does not draw.

8.5 Zoning layer evolution

The typed-zone classifier of Section 5 carries forward into v4 with several changes driven by using it on real generated output rather than only on held-out towns.

Leave-one-town-out macro-F1 for the zone classifier is approximately 0.30. This is a modest score, and the failure mode behind it is worth describing. The classifier tended to make an overconfident single-label call per pixel, and that call could swing between classes under small changes in the input calibration: the same growth pixel could be called residential under one density threshold and commercial under a neighbouring one. An overconfident classifier is more damaging in a planning display than an abstaining one, so v4 responds on several fronts rather than tuning the same classifier further.

Training-feature fidelity: the amenity feature used to train the classifier is now the sparse OSM amenity-point density used for labelling, not the model's own diffuse generated amenity channel: the two have different statistics, and training on the smoother generated channel while labels come from sparse points was teaching the classifier a feature distribution it would not see at inference. Margin-and-context abstention: a pixel whose classifier margin (top score minus runner-up) or whose local density context falls below a threshold is left untyped rather than forced into a class, so the over-confident-then-wrong failure mode shows up as "unlabelled" instead of a wrong colour on the map. A mixed-use class is added from proximity co-occurrence: OSM carries no first-class mixed-use landuse tag at this raster resolution, so a residential pixel near a commercial or retail pixel (and vice versa) is relabelled mixed-use in the training data, capturing live/work and ground-floor-retail patterns that a single-label residential-or-commercial split cannot. Residential density tiers are read directly from the generated density channel rather than re-derived, so the displayed residential intensity matches what the model actually generated instead of a separately fit threshold. A farmland class is added for rural context, so a growth pixel beyond the edge of town whose context reads as agricultural is assigned an appropriate class instead of being forced into one of the four urban classes. Finally, despeckling is unified: both the renderer and the classifier now share one connected-component filter that drops built-mask components below a small pixel-count threshold, so a real parcel and a single noisy pixel are no longer treated the same way by one stage and differently by the other.

8.6 Advisory zoning for expansion land

Until this point the zone classifier only touched pixels the diffusion model had already committed to: sample growth first, label it second. That ordering buries the most useful planning question. A planner deciding where a town should expand wants a recommended use for land before anything is built on it, not a label on a finished proposal.

v4 adds this as a second, explicitly advisory pass. The candidate set is the undeveloped ring within roughly 120 m of the current footprint, minus roads, water, steep slopes, and any protected mask. Each candidate pixel is typed with the same six features and the same classifier used for generated growth, with local density context taken from the existing town rather than from a sample. Pixels below the abstention margin stay untyped. On the map the advisory layer is drawn faint while sampled growth stays solid, so a proposal and a recommendation cannot be mistaken for one another.

The two layers answer different questions. The generated density says where growth is likely, given forty years of observed change on similar terrain. The advisory layer says what use fits a parcel if it develops. Where they agree, the planner gets a concrete suggestion: this fringe develops in most sampled futures, and its context reads as low-density residential with commercial frontage along the arterial. Where they disagree, the disagreement is itself information. The layer inherits every limitation of the classifier it wraps, including the 0.30 macro-F1, which is why it abstains rather than guesses and is presented as advice, not analysis.

8.7 Browser deployment

The diffusion model and the zone classifier both now run client-side. The trained U-Net is exported to a 23 MB ONNX file and run in-browser with onnxruntime-web, including the DDIM reverse-diffusion loop itself, so sampling a new future requires no server. The zone classifier is exported to JSON and its gradient-boosted decision trees are evaluated directly in JavaScript against the same feature stack used in Python; where we have checked it, this JavaScript evaluator reproduces the scikit-learn classifier's predictions exactly. A real-town mode fetches live elevation and OpenStreetMap data for the clicked location on demand, so the tool works on any town a user points at rather than only the towns in the training and evaluation lists, subject to the same OSM coverage caveats as the rest of the pipeline (Section 11).

9 v5: global expert coverage and amenity-scale correction

The v4 mixture of Section 8.3 encodes exactly two growth regimes, and most of the world's observed settlement growth belongs to neither. Informal peri-urban expansion around cities such as Lima, Nairobi, and Dhaka proceeds with building construction outrunning the mapped road network; the growth edges of megacities such as Delhi absorb population at densities the town and urban experts never see; and at the opposite end of the scale, village growth begins from windows that are almost entirely unbuilt. v5 addresses this coverage gap, together with a defect in the zone classifier's amenity feature that surfaced when the deployed tool was applied to a car-dependent United States suburb. We follow the same reporting convention as Section 8: the architecture, data, and correction are merged and released; training of the new experts had not completed at the time of writing, and no performance comparison is claimed.

9.1 A diagnosed failure: institutional over-prediction on sparse suburban fabric

Applied to a window over Middleburg Heights, Ohio (41.3584°N, 81.8039°W), the deployed zone classifier labelled 697 of 1,099 generated-growth cells and 4,869 of 11,345 advisory candidate cells as institutional, a majority and a plurality respectively, where the true institutional share of land use in such a suburb is a small fraction. The cause is traceable to a single design decision in the amenity-density feature. The feature is constructed by box-blurring OpenStreetMap point-of-interest counts and log-scaling the result, then dividing by the maximum value within the window. This per-window normalisation forces some location in every window to read 1.0. In the dense European and Asian training towns, where dozens of points of interest co-occur within a few blocks, a moderate normalised value is a reliable correlate of an institutional core. A car-dependent suburb inverts the pattern: a school or hospital standing alone on a large single-use parcel still becomes the window maximum, and the blur spreads an elevated halo around it. The classifier, never having observed an isolated, unclustered amenity peak, resolves the unfamiliar signal to the class most associated with amenity presence, which is institutional.

The correction has three parts. First, the feature moves from per-window to absolute calibration: the blurred, log-scaled density is divided by a fixed reference value, chosen empirically as the peak produced by a tight cluster of approximately twenty-five points of interest passed through the pipeline's own blur, and clipped to $[0, 1]$. Under this calibration an isolated point of interest reads approximately 0.07 rather than 1.0, and only a genuine cluster saturates the feature, so sparse and dense windows are measured on one axis. Because the amenity map is stored in the per-town caches and enters both the diffusion targets and the classifier features, the recalibration invalidates all caches and classifiers produced under the previous normalisation; the pipeline enforces separate cache directories for this reason. Second, the abstention mechanism of Section 8.5 is extended with per-class margins: a caller may require a larger classification margin for a designated class (institutional being the motivating case) without altering the treatment of the remaining classes, and the per-class threshold can only tighten the uniform one, never loosen it. Third, the classifier's training distribution now contains the offending morphology: the v5 window lists add Australian, New Zealand, and Canadian planned suburbs to the existing United States windows, so an isolated school on a parking lot is a pattern the classifier is trained on rather than an out-of-distribution surprise. A reproduction protocol accompanies the release: the pipeline re-runs advisory zoning on the Middleburg Heights window and on a dense held-out town under the previous normalisation (reconstructed as a control), under the absolute calibration, and under a sweep of institutional margins, reporting the class-count breakdown for each. Section 9.3 reports the outcome: the expanded training distribution alone removes the failure, and the recalibration's measurable effect on this window is small.

9.2 Five experts across the settlement scale

v5 widens the mixture to five experts sharing one architecture. The *village* expert trains on 22 windows around villages that grew measurably between 1980 and 2020, selected because the growth filter of Section 3.2 discards windows without observable change. The *town* expert retains the original hilly small-town data, and the *urban* expert retains the planned flat-fringe data of Section 8.3, extended by ten Australian, New Zealand, and Canadian suburban windows whose subdivision morphology matches the existing United States list. The *informal* expert trains on 24 unplanned peri-urban frontier windows across Latin America, Africa, and South and Southeast Asia. The *megacity* expert trains on 24 saturated growth-edge windows of the largest cities, including four in the Delhi region, with a fifth Delhi window (Narela) held out for evaluation alongside one held-out window per new expert. OpenStreetMap coverage is uneven across these lists; building mapping is thin in the two Chinese windows and road mapping is thin in several informal ones, and the growth filter drops windows whose data cannot support a training pair, at the cost of samples rather than correctness.

Routing remains a hard, ordered rule on conditioning statistics, for the reasons given in Section 8.3, with one added feature: the ratio of road-pixel fraction to built fraction. Informal frontiers put up buildings faster than mapped roads and planned fringes do the opposite, so the ratio separates the two regimes where built fraction alone cannot. The rule list is evaluated first match first: a built fraction below 0.02 routes to the village expert; above 0.22, to the megacity expert; a built fraction above 0.10 with a roads-per-built ratio below 0.35, to the informal expert; a mean slope above the v4 threshold, to the town expert; and the remaining flat, road-served windows to the urban expert. Scale therefore requires no additional mechanism: at a fixed 1.9 km window, the difference between a village and a Delhi growth edge is precisely how full and how road-served the window is, which is what the routing features measure. A window in central Delhi is already saturated and routes to the megacity expert; the growth question this model answers applies to the edges, which is where the held-out window sits. Because deployments rarely carry all five checkpoints, each expert declares an ordered fallback chain and the dispatcher routes to the first substitute actually loaded, logging both the requested and the substituted expert; a two-expert deployment reproduces the v4 behaviour exactly. The caveats of the v4 router carry over unchanged (a hard gate has no blending, and the thresholds encode our partition of the training data), joined by one new caveat: a window whose roads exist but are unmapped in OpenStreetMap is indistinguishable from an unplanned frontier in these statistics and will route to the informal expert. For routing purposes this is arguably the correct behaviour, since the model cannot condition on roads it cannot see, but the expert label should not be read as a claim about the settlement's legal or planning status.

9.3 Results

All five experts trained to 300 epochs on a single T4 GPU. Final noise-prediction losses were 0.024 (village), 0.028 (town), 0.032 (urban), 0.038 (informal), and 0.042 (megacity), ordered by the morphological complexity of the training windows; all five runs were stable, with no divergence. On the five held-out windows (Wengen, Gubbio, Des Moines, Antananarivo, Delhi-Narela) the router selected the intended expert in every case.

The zone classifier, retrained on the full v5 cache, reaches a leave-one-town-out macro-F1 of 0.257, against approximately 0.30 for the v4 classifier; transfer across five morphology families is a harder problem than transfer across hilly European towns alone. The retained classes are 1 through 5; no farmland labels survived the per-town sampling caps, so class 6 is wired through the pipeline but never assigned, and candidate farmland pixels abstain instead.

The reproduction protocol of Section 9.1 produced an instructive negative result. On the Middleburg Heights window, the control classifier (per-window maximum normalisation, retrained on the v5 cache) assigned 203 of 4,286 typed advisory cells to institutional, and the absolute-scale classifier assigned 221; both are near 5%, a plausible institutional share for such a suburb, and far from the 43% of the originally reported failure. Because both variants were trained on the expanded window lists, the comparison indicates that the training-distribution expansion, not the feature recalibration, removed the failure: once sparse suburban windows are present in training, the classifier handles an isolated school correctly under either normalisation, and the two variants produce identical macro-F1 to three decimal places. The recalibration is retained on measurement-consistency grounds, since it places sparse and dense windows on one axis, but we attribute the repair to the data. The per-class margin behaves as designed and monotonically: on the same window, institutional assignments fall from 221 to 104, 32, and 1 as the institutional margin rises through 0.5, 1.0, and 2.0, with the counts of every other class unchanged. A margin of 0.5 retains an institutional share of roughly 2.5%, closest to a plausible ground truth, and is the deployed default. The dense in-distribution check did not regress: Gubbio's advisory typing is identical under every variant.

The comparison this protocol cannot make is against the originally deployed v4 classifier, whose training cache predates the v5 window lists; reconstructing it would require rebuilding the v4 cache under the old normalisation. The 43% failure figure therefore rests on the original deployment observation, and the controlled result above isolates only the normalisation, not the training data, as a variable. No comparison between the five-expert mixture and a single mixed model has been run.

10 Applications: towards automated city-building agents

The sandbox mode is a decision-making layer for game automation: at each turn it maps (heightmap, current city) to a construction increment. A complete agent for a title such as Cities: Skylines 2 additionally needs perception (reading the game's heightmap and current road/zone state, available through its modding API) and actuation (executing placements, also exposed to mods). Because our model consumes and produces rasters, the integration surface is thin, and the plan-import mode of Section 6 already parses drawn or exported maps into conditioning. We do not implement the agent here; we note that raster-to-game-geometry conversion (vectorising road rasters into placeable splines) is the main engineering step, and the skeleton-to-graph extractor of Section 5 is most of it.

11 Limitations

Temporal road proxy. OSM has no road history, so epoch-t roads are present-day roads clipped to the epoch-t footprint (Section 3.2). Roads built recently inside an old footprint leak into early-epoch conditioning, making the conditioning network denser than the town actually was; the density channels are unaffected. **Density granularity.** GHS-BUILT-S is native $\sim$100 m (3 arc-second) resolution resampled to our 15 m/px grid, so the density target is smoother than the road raster and cannot carry parcel-level structure; the road channel and the 15 m DEM carry the fine detail. **Label sparsity outside Europe.** OSM amenity, landuse, and cycleway coverage thins outside Europe, so the scorecard's destination set is less complete in the Asian towns and the post-hoc classifiers are trained on European labels; leave-one-town-out transfer is tested only within that labelled pool. The v5 window lists extend this concern: road mapping is thin in several informal-frontier windows and building mapping is thin in the Chinese megacity windows, so the growth filter drops some of exactly the windows those experts most need, and the surviving informal windows over-represent the better-mapped frontiers. **Window scale.** 1.9 km windows exclude transit-scale structure, such as tram and rail corridors, which is why transit appears in the scorecard only as a destination category, not as generated infrastructure. **Scorecard scope.** The eleven metrics are spatial-form proxies, not certified environmental performance (Section 4); no zoning law, land ownership, budget, or hydraulic model enters the pipeline, and generated layouts are proposals, not plans. **Baselines.** The slope-blind control is deliberately weak; comparison against SLEUTH-class CA models and layout generators retrofitted with terrain channels is future work. **Road channel unlearnable at this data scale.** Even after the mixed-dataset retrain of Section 8.1, the 1-pixel-wide road target does not reach the spatial coherence the density and amenity channels reach (neighbour correlation $\sim$0.26 against $\sim$0.98); we hide the generated road channel in deployment rather than present an incoherent one, and identify road-target dilation as the untested fix. **Water conditioning not yet retrained.** The fifth conditioning channel of Section 8.2 is implemented and wired into the data pipeline, but the water-conditioned checkpoint has not finished training; the RePaint water/steep-slope lock is a deployment patch on the existing 4-channel model, not evidence the conditioning channel is unnecessary. **No mixture-versus-single-model comparison.** All five experts of Section 9.2 are trained, but no controlled comparison between the mixture and a single model trained on the pooled data has been run, for v5 or for the v4 two-expert design; the case for the mixture rests on the training-distribution argument, not on measured output quality. **Router thresholds are a data partition.** The five-way routing thresholds of Section 9.2 were chosen from the training-window distributions, not learned or validated against an external definition of settlement type; a window whose roads are unmapped in OSM routes as informal regardless of its actual planning status. **Amenity recalibration invalidates earlier artefacts.** The absolute amenity scale of Section 9.1 changes both a diffusion target channel and a classifier feature, so caches, checkpoints, and zone classifiers produced before the change cannot be mixed with v5 artefacts; cross-version comparisons require retraining the control, which the released protocol does. **Zoning classifier accuracy.** Leave-one-town-out macro-F1 for the typed-zone classifier is 0.257 after the v5 retrain, below the $\sim$0.30 of the v4 classifier, reflecting the harder transfer problem across five morphology families; abstention reduces the cost of errors but does not raise the underlying accuracy, and the classifier's output should be read as indicative rather than authoritative. The amenity recalibration did not change macro-F1, consistent with the small weight the feature carries in the trees. **Farmland class inactive.** No farmland labels survived the per-town sampling caps in the v5 cache, so the farmland class exists in the pipeline but is never assigned; rural growth pixels abstain instead of receiving an agricultural type. **Planner-tooling scope.** The natural-language operations layer of Section 8.4 covers a fixed, closed set of operations and a fixed region vocabulary; requests outside that vocabulary produce a warning, not a best-effort guess, so the tool's conversational range is narrower than free-form natural language would suggest.

12 Conclusion

We presented a terrain-conditioned diffusion model for street-level growth of small towns, trained on real 1980–2020 built-up change assembled fully automatically from open data, scored and selected by an eleven-metric sustainability scorecard, annotated post hoc with bike infrastructure and typed zones, and usable as a proposal tool for real settlements, an iterative growth engine on arbitrary heightmaps, and a continuation engine for human-drawn plans. The v4 retrain on a larger, mixed dataset shows the density and amenity channels learning coherent, conditioning-faithful output while the road channel does not yet learn at this data scale, a result we report plainly rather than obscure. A fifth conditioning channel for water, a two-expert mixture with an auditable router, and a planner-facing layer of region-locked inpainting, consensus maps, vector export, and validated natural-language operations move the pipeline further toward a tool a planner could actually use. v5 widens the mixture to five experts spanning village, hilly town, planned fringe, informal frontier, and megacity edge, with 84 new windows across five continents and a Delhi-region hold-out, and corrects a diagnosed amenity-normalisation defect that had caused institutional over-prediction on sparse suburban fabric. The five training runs and the before/after validation are complete; the validation attributes the repair to the expanded training data rather than the feature recalibration, a distinction the deployment retains in the form of per-class abstention margins. A controlled mixture-versus-single-model comparison remains open.

AI assistance disclosure

The code, experiments, and text of this paper were produced with substantial assistance from an AI system (Anthropic Claude), directed and reviewed by the author. This disclosure is made in accordance with venue policies on AI-assisted research.

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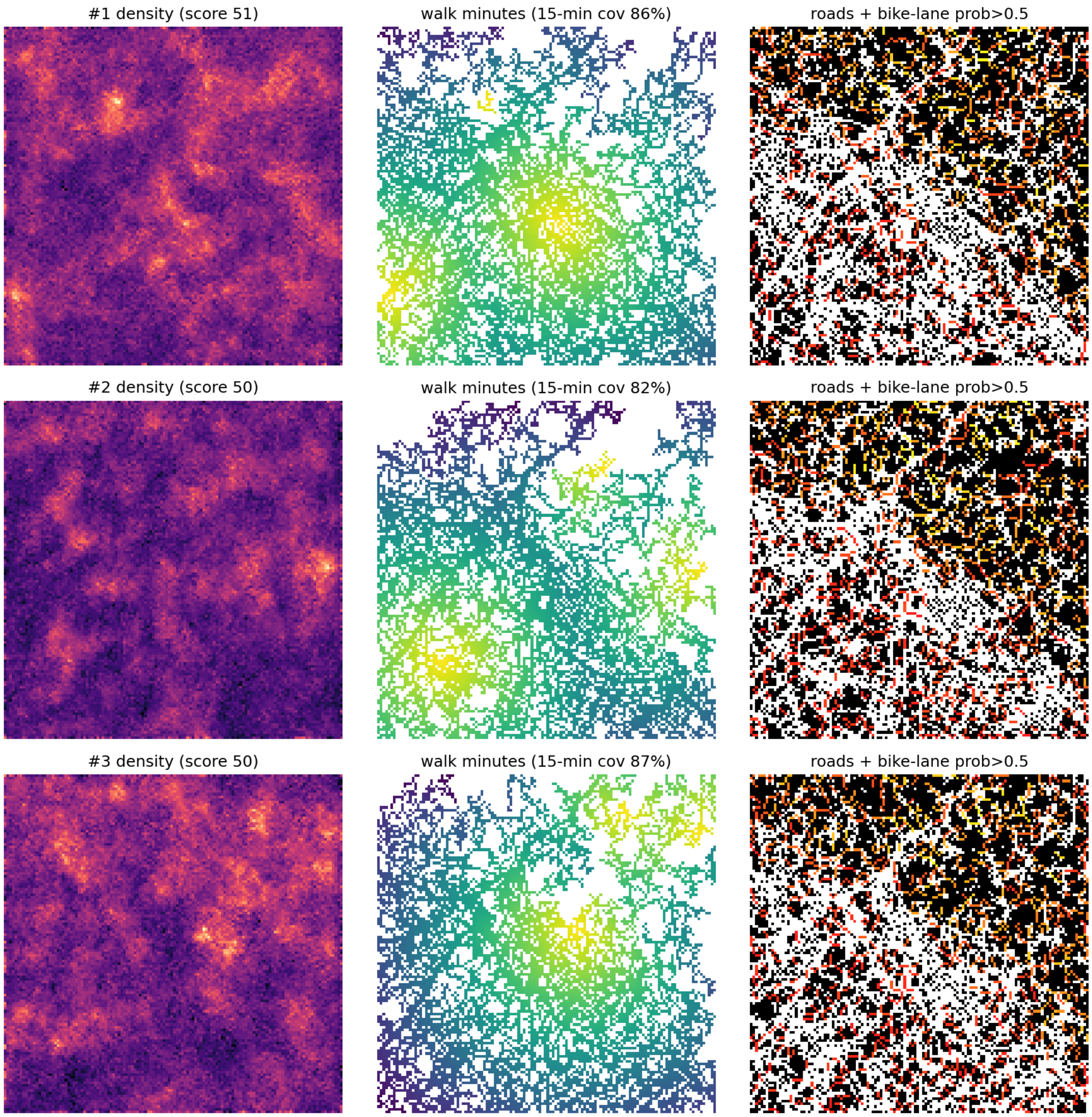

Figure 2: Top-3 of N sampled futures for a held-out town, ranked by the eleven-metric scorecard (rows; total scores 51, 50, 50 of 100). Left: generated continuous built-density channel. Centre: per-pixel walk time along the generated road network to the nearest real OSM amenity (Dijkstra at 80 m/min), with the share of built pixels inside the 15-minute budget given per sample (86%, 82%, 87%). Right: road pixels (black) with graph edges whose predicted bike-infrastructure probability exceeds 0.5 overlaid in warm colours by the post-hoc classifier of Section 5. These are raw model rasters, deliberately shown without the plan-style rendering stage.